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Course outline

About NPTEL

How does an NPTEL online course work?

Week 1

Week 2

- Introduction to Anaconda
- Installation of Anaconda
- Introduction to Spyder IDE
- Understanding Variables in Python
- Executing a sequence of instructions in the Console
- Printing statements in Python
- Writing your First Program
- Taking inputs from the user
- Discount Calculation
- Motivation of if condition
- A reminder on how to deal with numbers
- Understanding if condition's working
- Realizing the importance of syntax and indentation
- Introduction to loops
- Loops: Sum of numbers
- Loops: Sum of numbers (continued)
- Logic: Multiplication Tables
- Introduction to While loop
- Week 2 Feedback Form: The Joy of Computing using Python
- Quiz: Week 2 : Assignment 2
- Week 2 - Programming Assignment 1
- Week 2 - Programming Assignment 2
- Week 2 - Programming Assignment 3
- Week 2 solution

Week 3

week 4

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

Text Transcripts

Download Videos

Problem Solving Session - July 2025

Week 2 : Assignment 2

The due date for submitting this assignment has passed.

Due on 2025-08-06, 23:59 IST.

As per our records you have not submitted this assignment.

Ria is building a Python assistant for her family's retail store. The assistant should help calculate discounts on items based on the amount the customer spends.

If the total amount is:

- More than ₹5000 → 20% discount
- Between ₹3000 and ₹5000 → 10% discount
- Less than ₹3000 → No discount

She also wants the program to **take customer input, calculate final price, and print the savings.**

1) What is the most appropriate data type to take customer input, calculated final price, and print the savings? *1 point*

- int
- float
- bool
- str

No, the answer is incorrect.
Score: 0
Accepted Answers:
float

2) What is the correct Python syntax to take customer purchase amount as input? *1 point*

- input(float())
- float(input())
- input = float()
- float = input()

No, the answer is incorrect.
Score: 0
Accepted Answers:
float(input())

3) **Which of the following correctly applies a 20% discount on the purchase amount stored in a variable amount?** *1 point*

- discounted = amount * 0.2
- discounted = amount - amount * 0.2
- discounted = amount - 20
- amount = amount * 1.2

No, the answer is incorrect.
Score: 0
Accepted Answers:
*discounted = amount - amount * 0.2*

4) **If a customer spends more than ₹5000, they get a 20% discount. Which of the following is the correct if statement to check this?** *1 point*

- if amount > 5000:
- if amount => 5000:
- if amount = 5000:
- if amount >= 5000:

No, the answer is incorrect.
Score: 0
Accepted Answers:
if amount > 5000:

5) What will be the output of the following code if the amount entered is 5000? *1 point*

```
amount = 5000
if amount > 5000:
    amount = amount - (amount * 0.2)
print(amount)
```

- 4000
- 4500
- 5000
- 3000

No, the answer is incorrect.
Score: 0
Accepted Answers:
5000

6) **Ria wants to show the customer how much they saved. Which line of code correctly prints both the original price and the savings using f-strings?** *1 point*

- print("Original: ₹", amount, "Saved: ₹", saved)
- print(f"Original: ₹{amount}, Saved: ₹{saved}")
- print("Original: ₹amount, Saved: ₹saved")
- print(f"Original: ₹amount, Saved: ₹saved")

No, the answer is incorrect.
Score: 0
Accepted Answers:
print(f"Original: ₹{amount}, Saved: ₹{saved}")

7) Identify the error in the following code: *1 point*

```
amount = float(input("Enter amount: "))
if amount > 3000
    print("You got 10% off!")
```

- Missing else block
- Input cannot be used with float
- Colon is missing after if condition
- float should be int

No, the answer is incorrect.
Score: 0
Accepted Answers:
Colon is missing after if condition

8) **What is the final price after applying the following logic?** *1 point*

```
amount = 6000
if amount > 5000:
    amount = amount * 0.8
elif amount > 3000:
    amount = amount * 0.9
else:
    pass
```

- 6000
- 4800
- 5400
- 3000

No, the answer is incorrect.
Score: 0
Accepted Answers:
4800

9) **Which of the following best implements nested if logic to apply 20% discount if amount > 5000 and is a loyalty customer? Assume is_loyal = True** *1 point*

```
if amount > 5000 and is_loyal:
    amount = amount * 0.8
```

```
if amount > 5000:
    if is_loyal:
        amount = amount * 0.8
```

```
if is_loyal:
    amount = amount * 0.8
```

```
if amount > 5000:
    if is_loyal:
        amount = amount * 0.8
```

No, the answer is incorrect.
Score: 0
Accepted Answers:

```
if amount > 5000:
    if is_loyal:
        amount = amount * 0.8
```

10) **What will be the output of the following program if the user inputs 6000?** *1 point*

```
amount = float(input("Enter amount: "))
discount = 0

if amount > 5000:
    discount = 0.2
elif amount > 3000:
    discount = 0.1

final_price = amount - amount * discount
print("Final Price:", final_price)
```

- **Final Price: 4800.0**
- **Final Price: 4400.0**
- **Final Price: 5400.0**
- **Final Price: 6000.0**

No, the answer is incorrect.
Score: 0
Accepted Answers:
Final Price: 4800.0

Ravi is a school teacher creating a Python-based multiplication table trainer for his students. The program should: Ask the student for a number as input. Use a loop to print the multiplication table of that number up to 10. Extend it to practice sessions where the program keeps asking for new numbers until the user enters 0. At the end, print a summary of how many tables were practiced.

11) **Which Python loop is most suitable to print the multiplication table of a number from 1 to 10?** *1 point*

- while loop
- for loop
- if loop
- do while loop

No, the answer is incorrect.
Score: 0
Accepted Answers:
for loop

12) **Ravi writes the following code to print the multiplication table of 5. What will be the output of the first line?** *1 point*

```
num = 5
for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

- 5 x 0 = 0
- 5 x 1 = 5
- 1 x 5 = 5
- 5 x 1 = 55

No, the answer is incorrect.
Score: 0
Accepted Answers:
5 x 1 = 5

13) **What will be the output of range(1, 11) inside a for loop?** *1 point*

- 1 to 10 (inclusive)
- 0 to 10 (inclusive)
- 1 to 11 (inclusive)
- 1 to 9 (inclusive)

No, the answer is incorrect.
Score: 0
Accepted Answers:
1 to 10 (inclusive)

14) **Which of the following loops will correctly continue asking the user for a number until they enter 0?** *1 point*

```
while num != 0:
    num = int(input())
```

```
while True:
    num = int(input())
    if num == 0:
        break
```

```
if num != 0:
    num = input()
```

```
for num in range(10):
    input()
```

No, the answer is incorrect.
Score: 0
Accepted Answers:

```
while True:
    num = int(input())
    if num == 0:
        break
```

15) What is the total number of lines printed by the following code ? *1 point*

```
num = 7
for i in range(1, 11):
    print(num, "*", i, "=", num * i)
```

- 7
- 10
- 11
- 0

No, the answer is incorrect.
Score: 0
Accepted Answers:
10

16) Ravi wants to track how many tables were generated before the student exited. Which of the following variables and logic should be used? *1 point*

- Use count = 0, increment count by 1 inside the loop each time a table is printed
- Use count = 1, decrement each time
- Use count = 10, multiply inside the loop
- No counter is needed

No, the answer is incorrect.
Score: 0
Accepted Answers:
Use count = 0, increment count by 1 inside the loop each time a table is printed

17) What will be the output of the following code? *1 point*

```
num = 3
i = 1
while i <= 5:
    print(num * i)
```

- Prints multiples of 3 up to 15
- Prints an error
- Infinite loop
- Prints nothing

No, the answer is incorrect.
Score: 0
Accepted Answers:
Infinite loop

18) Ravi wants the output to look like: 5 x 1 = 5 but accidentally writes: *1 point*

```
print(i, "x", num, "=", i * num)
```

For input num = 5, i = 1, what will be printed?

- 5 x 1 = 5
- 1 x 5 = 5
- x 5 = 5
- i x num = result

No, the answer is incorrect.
Score: 0
Accepted Answers:
1 x 5 = 5

19) **Which keyword is used to skip the current iteration of a loop and continue with the next one?** *1 point*

- stop
- break
- continue
- pass

No, the answer is incorrect.
Score: 0
Accepted Answers:
continue

20) What will this code output if the user enters the following numbers sequentially in the terminal: 3, 4, 5, 0? *1 point*

```
count = 0
while True:
    num = int(input("Enter number: "))
    if num == 0:
        break
    for i in range(1, 11):
        print(num * i)
    count += 1
print("You practiced", count, "tables.")
```

- 3 tables printed, followed by: "You practiced 3 tables."
- Infinite loop
- Error due to missing condition
- 4 tables printed

No, the answer is incorrect.
Score: 0
Accepted Answers:
3 tables printed, followed by: "You practiced 3 tables."

In the quiet village of **Phulera**, the computer in the **Panchayat office** starts displaying strange numbers. **Abhishek Sir,** the village secretary, thinks it might be a software glitch. But his close friends **Vikas** and **Prahlad** suspect something else — maybe it's a secret code!

They notice the numbers follow some strange patterns:

. **Armstrong numbers** (like 153)

. **Neon numbers** (like 9 → 9² = 81 → 8 + 1 = 9)

. **Palindrome numbers** (like 121)

So, the team writes a Python program to check if a number is an **Armstrong number** — one of the patterns they suspect is being used.

21) *1 point*

```
num = int(input("Enter a number: "))
original = num
sum = 0
count = 0
temp = num
while temp > 0:
    count += 1
    temp = temp // 10
temp = num
while temp > 0:
    digit = temp % 10
    sum += digit ** count
    temp = temp // 10

if sum == original:
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

What is the role of the first while loop in the program?

- It checks whether the number is divisible by 10
- It calculates the cube of digits
- It finds how many digits are in the number
- It stores the reverse of the number

No, the answer is incorrect.
Score: 0
Accepted Answers:
It finds how many digits are in the number

22) **What will be the output when the input is 9474?** *1 point*

- Armstrong Number
- Not an Armstrong Number
- Error due to integer overflow
- Infinite loop

No, the answer is incorrect.
Score: 0
Accepted Answers:
Armstrong Number

23) **How many times does the body of each while loop execute when the input is 1000?** *1 point*

- 3
- 4
- 5
- It never executes

No, the answer is incorrect.
Score: 0
Accepted Answers:
4

24) **What is the value of sum immediately after the second while loop finishes when the input is 370?** *1 point*

- 0
- 370
- 3700
- 27

No, the answer is incorrect.
Score: 0
Accepted Answers:
370

25) **If the line temp = num that appears just before the second loop is deleted, what will the program output for input 9474?** *1 point*

- Armstrong Number
- Not an Armstrong Number
- The program hangs
- Raises NameError

No, the answer is incorrect.
Score: 0
Accepted Answers:
Not an Armstrong Number

26) **During execution with input 1000, what is the value of digit in the third iteration of the second while loop?** *1 point*

- 0
- 1
- 10
- The loop has no third iteration

No, the answer is incorrect.
Score: 0
Accepted Answers:
0

27) **Which replacement for the line if sum == original: will make the program print "Armstrong Number" for every positive input but "Not an Armstrong Number" for zero?** *1 point*

- [ ] if sum >= original:
- [ ] if original != 0:
- [ ] if original > 0:
- [ ] if True:

No, the answer is incorrect.
Score: 0
Accepted Answers:
if original != 0:
if original > 0:

28) **What final value does count take when the input is 1634?** *1 point*

- 3
- 4
- 5
- 6

No, the answer is incorrect.
Score: 0
Accepted Answers:
4

29) **Which single-line change will guarantee that no input at all (including 0) is reported as an Armstrong number?** *1 point*

- Initialize sum to 1 instead of 0
- Initialize count to 1 instead of 0
- Replace digit = temp % 10 with digit = abs(temp % 10)
- Change the condition to if sum >= original:

No, the answer is incorrect.
Score: 0
Accepted Answers:
Initialize sum to 1 instead of 0

30) **What does the program print when the user enters -153?** *1 point*

- Armstrong Number
- Not an Armstrong Number
- Infinite loop
- Raises ValueError

No, the answer is incorrect.
Score: 0
Accepted Answers:
Not an Armstrong Number